

Forensic Microscopy

Lab 6: Biaxial Interference figures

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Where can you find the slides?



- <https://aritra-evolves.github.io/FIVS215.html>
- Canvas

(Grab a Microscope from the black closet, set up kohler)

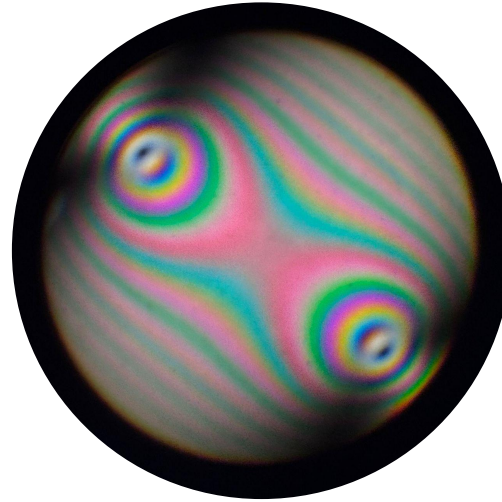
Today's Agenda

- ❖ Biaxial vs. Uniaxial
- ❖ Viewing
- ❖ Practice

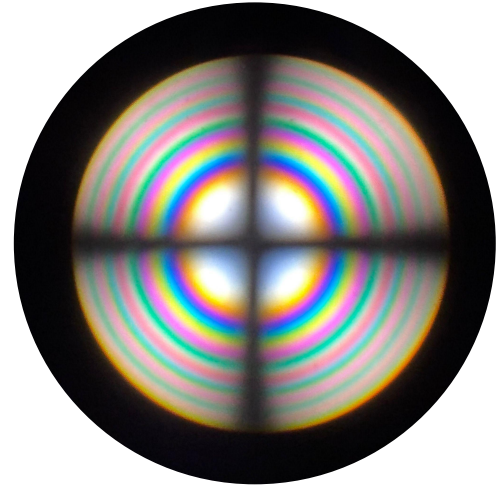


Biaxial vs. Uniaxial

- Roman cross vs. Hyperbolas
- Determine by rotating the stage
- Biaxial materials are either orthorhombic, monoclinic, or triclinic crystal lattice
- Best observed with mylar
- Excalibur



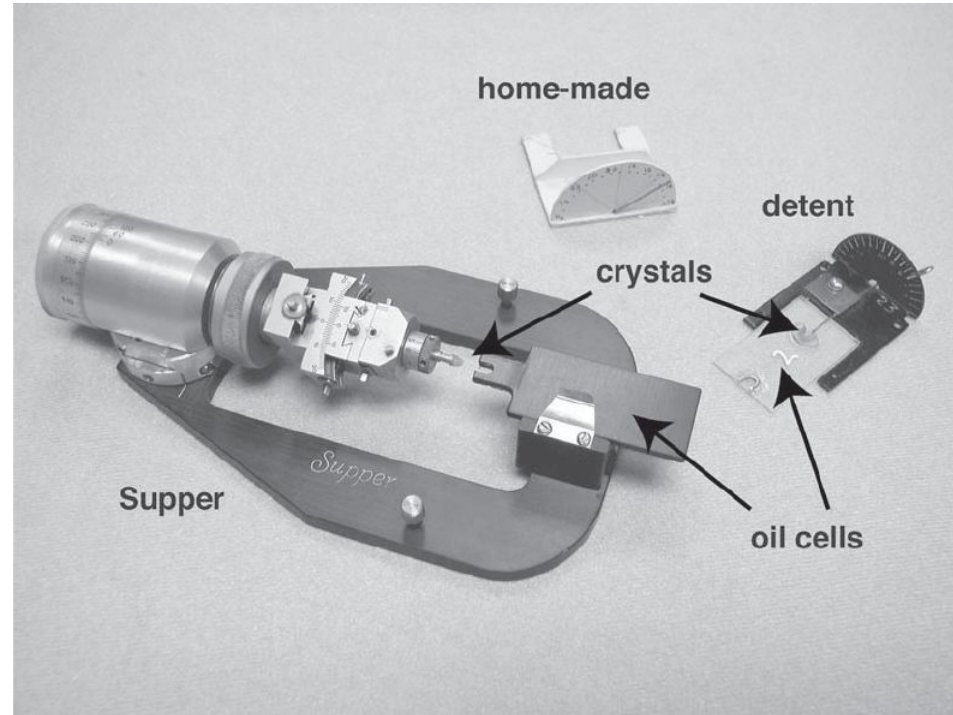
Biaxial



Uniaxial

Rotating the Crystal

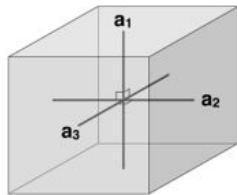
- Goniometer, spindle stage, or crystal rolling
- Spindle stage is the cheapest option
 - It's the cheap version of a goniometer
- When would this be useful?



Isotropic

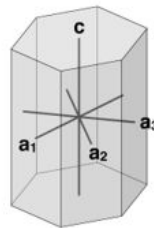
Uniaxial

Isometric



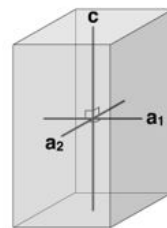
$$a_1 = a_2 = a_3;$$
$$a_1 \perp a_2 \perp a_3$$

Hexagonal



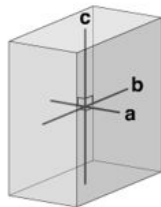
$$c \neq a_1 = a_2 = a_3;$$
$$c \perp a_1, a_2, a_3; \angle_{a \text{ axes}} = 60^\circ$$

Tetragonal



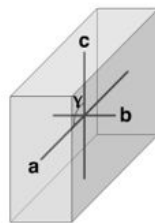
$$c \neq a_2 = a_3;$$
$$a_1 \perp a_2 \perp a_3$$

Orthorhombic



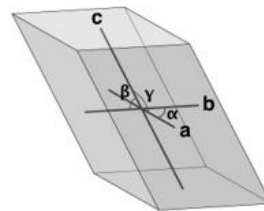
$$c \neq b \neq a;$$
$$c \perp b \perp a$$

Monoclinic



$$c \neq b \neq a;$$
$$c \perp b; b \perp a; \gamma \neq 90^\circ$$

Triclinic

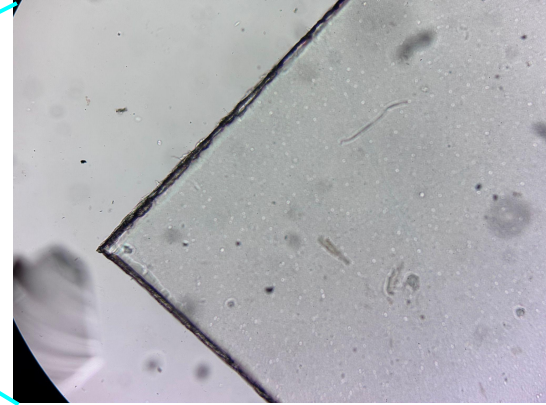
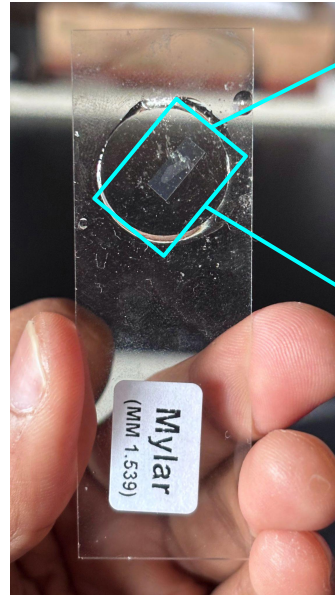


$$c \neq b \neq a;$$
$$\alpha, \beta, \gamma \neq 90^\circ$$

Biaxial

Practice

1. Get a Mylar Sample from the Back
2. Pull Out your Wave Plate
3. Set up Kohler on 10x
4. Cross your Polars
5. Look for Areas that Remain Gray when Rotating
6. Swing in a Higher Objective
7. Open your Substage Aperture
8. Swing in the Bertrand Lens
9. Observe the Roman Cross or Hyperbolas
 - a. Rotate the stage so the circles are in II & IV
10. Insert a Compensator or Quartz wedge to Observe the Optic Sign

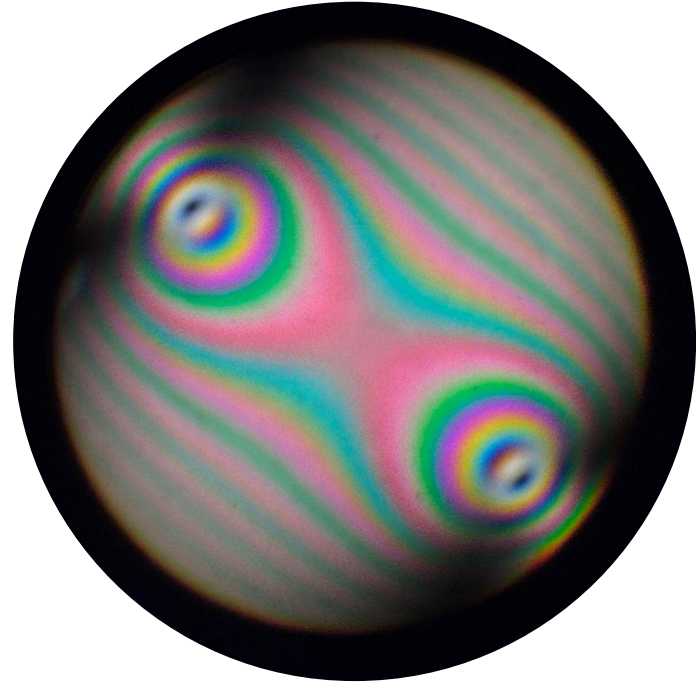


Make sure you're looking at the right thing.

Steps 6-9 – What should you see?

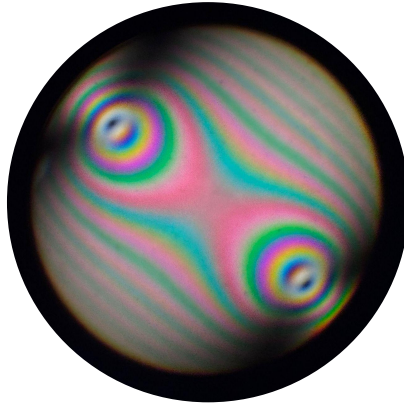
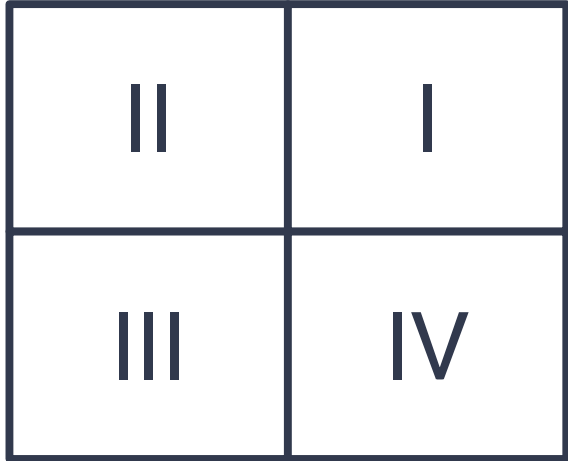
- A figure eight of sorts
- Rainbow of colors
- Make sure you've rotated the stage so that the circles are in II & IV

II	I
III	IV



Slow axis and fast axis

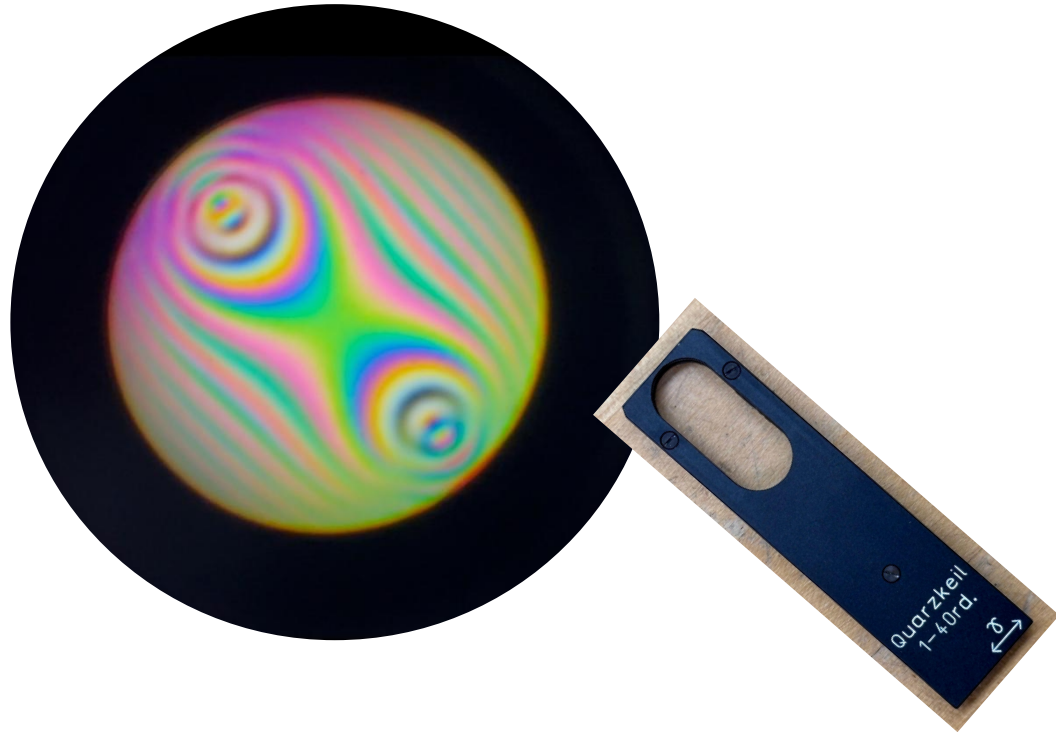
- **Fast Axis (α'):** The direction where the light ray encounters the **lower refractive index** (n), allowing it to travel faster.
- **Slow Axis (γ'):** The direction where the light ray encounters the **higher refractive index** (n), causing it to travel slower.



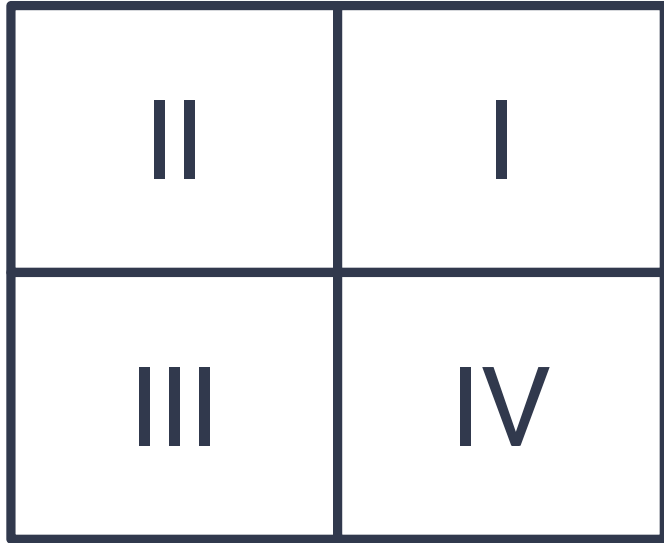
The compensator plates usually mark their slow axis

Add in a Quartz Wedge

- Add in the Quartz Wedge
- You should see the colors move in or out in waves
- Keep track of which way they move



Positive vs. Negative: $\epsilon - \omega = \text{optic sign}$



- Observe how the colors move in I & III
- If the colors move OUT
 - Epsilon is PARALLEL to the slow axis
 - Sign is **NEGATIVE**
- If the colors move IN
 - Epsilon is PERPENDICULAR to the slow axis
 - Sign is **POSITIVE**

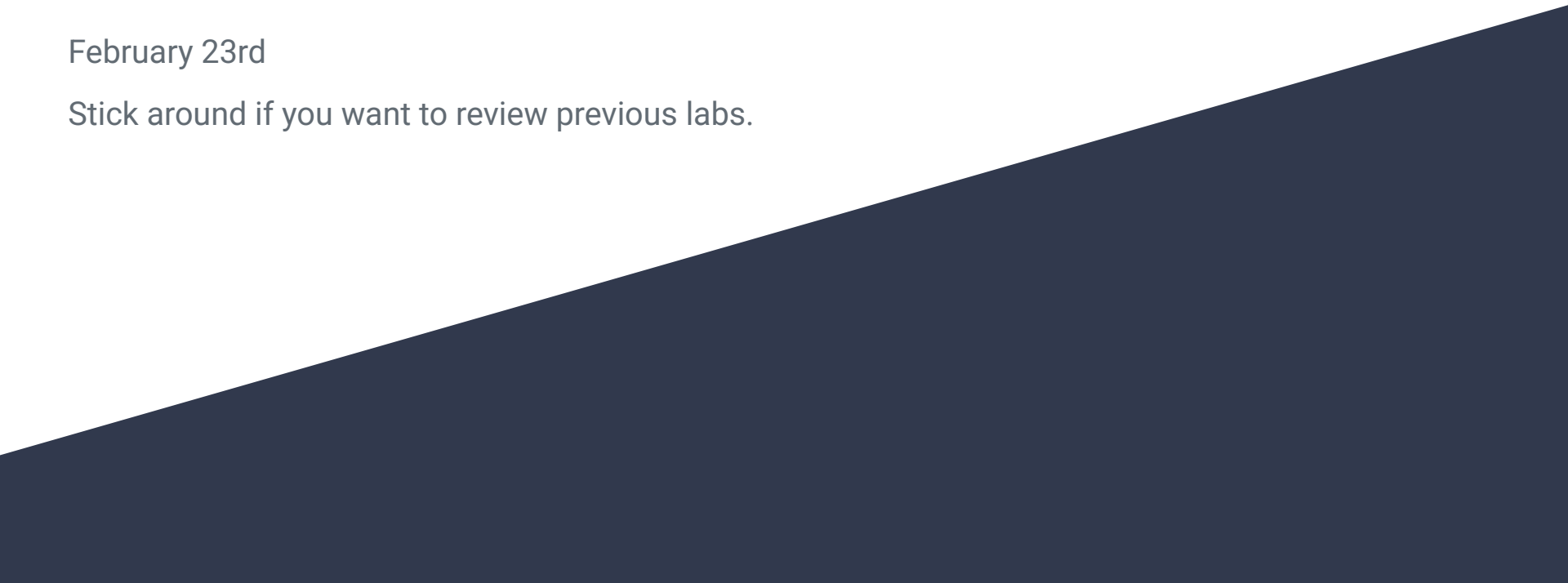
Michel Levy Chart



See you next week!

February 23rd

Stick around if you want to review previous labs.

A dark blue diagonal gradient bar that starts from the bottom left corner and extends towards the top right corner, covering the lower half of the slide.